Robust and Discriminative Visual-Semantic Alignment for Zero-Shot Remote Sensing Image Scene Classification

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***Abstract*— With the rapid development of remote sensing (RS) technology, the range of its applications, such as urban planning and environmental monitoring, has broadened considerably. However, the vast increase in RS data poses challenges for training deep neural networks (DNNs), primarily due to the extensive labeled datasets required. To address this issue, zero-shot learning (ZSL) emerges as a viable approach, enabling recognition of unseen categories without relying on training samples. In this article, we introduce a robust and discriminative visual-semantic alignment (RDVSA) model, which utilizes attribute-guided contrastive learning for enhanced performance. By incorporating category centroids into the embedding space of visual features and attribute descriptions, our model enhances the discriminative power of learned features and stabilizes the learning process, particularly in the presence of complex data distributions and noise in RS images. Additionally, we employ a vision transformer (ViT) and a "mean-teacher" framework to capture both global context and local details, ensuring robustness in occluded or irrelevant attribute regions. Comprehensive experiments on large-scale RS benchmarks reveal that our approach surpasses current state-of-the-art methods in both ZSL and generalized ZSL (GZSL) scenarios. These results underscore the effectiveness of using attribute-based contrastive learning alongside consistency constraints.**

***Index Terms*— Contrastive learning, remote sensing (RS) scene classification, visual-semantic alignment, zero-shot learning (ZSL).**

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I. Introduction

With the rapid progress in sensor technology and remote sensing (RS), classification of RS imagery has become essential for various applications, including urban construction [1] and environmental monitoring [2]. Deep neural networks (DNNs) have achieved notable success in this field [3], but their performance is contingent on access to substantial datasets, which are generated by Earth observation systems [3], [4]. The expansion of RS databases has increased the urgency of recognizing new and previously unseen scenes. Zero-shot learning (ZSL) [5] offers a solution to this challenge by enabling the recognition of novel categories in the absence of training samples, a strategy that has been investigated in the context of image classification [6] and object detection [7].

As illustrated in Fig. 1, ZSL models are capable of extending acquired knowledge to new classes by transferring semantic knowledge alongside visual features from each observed class. ZSL relies on two components: class embeddings, which use semantic descriptors like attributes [8] or word vectors [9], and projection functions that map visual features into this shared space [10], enabling transfer to unseen classes. The enhancement of class embeddings typically hinges on two principal approaches: the use of manually annotated attributes or the deployment of semantic embeddings derived from pretrained language models. Attributes, which encapsulate object features [11], are extensively utilized and have proven to be highly effective in ZSL tasks [5], [10], [12]. In recent years, there has been a growing interest in leveraging pretrained language models, such as word2vec and BERT, to extract semantic embeddings as substitutes for manual attributes. Furthermore, a study [13] has established a novel RS knowledge graph (RSKG), which enables the generation of semantic representations for scene categories through the representation learning capabilities of the RSKG. To mitigate the expenses associated with manually annotating RS imagery, Xu et al. [14] pioneered the incorporation of attribute-based semantic representations into the field of RS. They proposed the refinement of the CLIP model [8] to autonomously gather attributes that are perceptible through visual inspection.

Building upon attribute annotation techniques, many ZSL methods learn compatibility functions to align visual and attribute representations. In generalized ZSL (GZSL), visual features are projected into a semantic space via embedding

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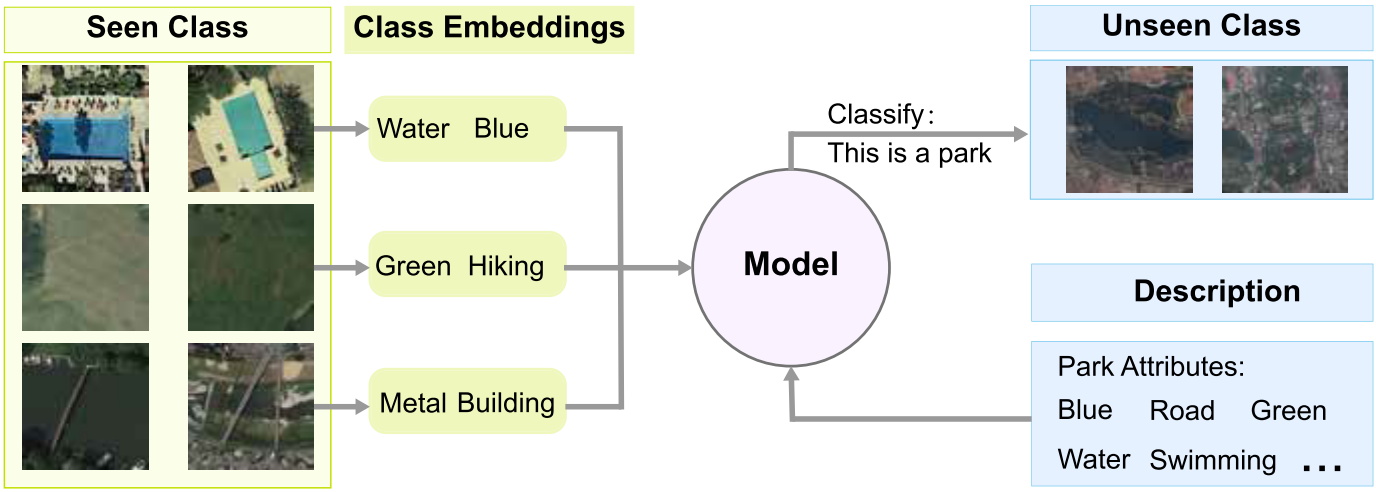


Fig. 1. Illustration of ZSL. The model learns semantic attributes water, green, and metal from seen classes swimming pool, grass, and bridge and then recognizes the unseen class park by combining attributes water, road, and green. This alignment bridges visual and semantic spaces for classifying novel scenes.

functions [10], [15], [16], where instance embeddings align with category-level attribute descriptions. Existing methods often establish correspondences by linking image regions to visual attributes [10], [17], [18], with attention mechanisms or attribute classifiers [19] enhancing localization. For instance, semantic vectors guide attention-based methods [20] to identify relevant visual attributes for local feature extraction. However, these methods often focus on regional embeddings but neglect their discriminative potential. Unlike natural images, RS scenes lack explicit foreground objects and exhibit intraclass diversity such as color variations in sea scenes under environmental conditions and interclass confusion exemplified by mobile home park versus dense residential areas. The DSVA framework [14] pioneered CLIP-based attribute annotation and ViT for global–local feature fusion, but struggles with these complex RS challenges due to unstable contrastive sampling and limited robustness to noise. To address these limitations and bolster the stability of ZSL in classifying RS images, we propose a robust and discriminative visual-semantic alignment (RDVSA) model. In this context, robustness is defined as the model's ability to maintain stable classification performance under challenging conditions such as unseen categories, occlusions, or noisy data in RS scenes. Discriminative capacity refers to the model's capability to learn separable feature representations that effectively distinguish between visually similar categories, such as mobile home parks and dense residential areas. Our framework employs a ViT to integrate global context and local details through its self-attention mechanism, enabling robust feature learning in complex RS scenes. Building on this foundation, RDVSA proposes centroid-based contrastive learning (CBCL) to replace DSVA's unstable random sampling by clustering samples around class centroids. Additionally, a mean-teacher framework is incorporated, where a teacher network guides a student network through consistency regularization under varying image perturbations. In summary, this article presents three pivotal contributions.

1) We propose contrastive embeddings that incorporate class centroids into the embedding space of visual features and attribute descriptions. The robustness of centroid samples helps the embedding space form clearer category boundaries, and the weakly supervised representation learning framework guided by attributes enhances the discriminative power of the learned visual features.
2) The proposed model addresses the complex data distribution and high noise in large-scale RS scenes by combining a ViT with a mean-teacher framework. This design enhances global–local feature learning while enforcing consistency across transformed inputs, resulting in improved robustness and generalization compared to existing ViT-based ZSL approaches.
3) In both ZSL and GZSL settings, our model was extensively tested on large-scale RS scene benchmarks, demonstrating that our method outperforms most state-of-the-art methods, validating the effectiveness of our approach.

II. Related Work

A. ZSL for RS Scene Classification

In ZSL, the objective is to train a model with samples from classes that are visible and then discern unseen classes by capitalizing on class embeddings or what is also referred to as ancillary information [5]. One of the pioneering methods in ZSL was put forth in [21], which pioneered the concept of attribute prediction for classes not encountered during training. Since its inception, ZSL methodologies have diverged into two principal streams. The first, epitomized by ALE [12], concentrates on aligning image representations with class embeddings and honing the compatibility function that exists between them [14], [22]. The second category embraces generative models to fabricate image features for unseen classes, leveraging generative adversarial networks (GANs) [23] and variational autoencoders (VAEs) [24] that are conditioned on these class embeddings.

In the RS field, early endeavors include the work of [25], who employed unsupervised domain adaptation and label propagation techniques, and [26], who crafted a bilinear compatibility function. LPDCMENs [27] advanced the state of

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the art by training an end-to-end network that underscores pairwise similarity within RS imagery. Subsequently, Li et al. [28] introduced the DAN model, leveraging generative methods to harmonize visual and semantic features, while [9] proposed the DSAE autoencoder with Euclidean constraints. However, the distinctive attributes of RS images, such as pronounced interclass variability and the absence of a distinct foreground–background demarcation, constrain the applicability of these models. To surmount these challenges, we harness a vision transformer (ViT) equipped with a self-attention mechanism to apprehend both local and global contexts within RS images, augmented by an attention module designed to sharpen interclass discrimination.

B. Contrastive Learning Strategies for ZSL

Contrastive learning is an effective strategy for constructing a cohesive embedding space, where samples from the same class are clustered closely together and those from different classes are positioned further apart. This is achieved by amplifying the similarity between a sample and its positive pairs—typically, other instances of the same class—while diminishing the similarity with negative pairs, which are samples belonging to different classes. This approach is applicable in both supervised and unsupervised learning scenarios. Milestones in contrastive learning include techniques designed to maximize mutual information across various representations of a single image [29], methodologies for crafting effective negative samples to bolster visual representation learning [30], and the development of the MoCo model, which underscores the benefits of leveraging a more extensive pool of negative examples [31].

Within the realm of ZSL, contrastive embedding techniques have been explored to bolster the capacity to differentiate between visual and semantic features. For instance, Jiang et al. [32] presents a methodology for transferring visual-semantic information from classes that are seen to those that are unseen by contrasting images across different classes. In a similar vein, Liu et al. [33] integrate a feature generation model with contrastive embeddings specifically designed for GZSL. Expanding upon these methodologies, our approach incorporates class-specific supervision to refine a robust attribute alignment model for ZSL, with the objective of enhancing the precision and dependability of attribute recognition for unseen classes.

C. Class Embeddings for ZSL

Class embeddings play a pivotal role in ZSL, facilitating the transfer of knowledge from seen to unseen classes through the exploitation of shared semantic attributes. These embeddings are generally categorized into three main types. The first type comprises human-annotated attributes, as elaborated in [11] and [34]. While these attributes are adept at capturing category-specific details, their compilation is laborious and time-consuming, a limitation highlighted by [6]. The second type encompasses embeddings derived from pretrained language models, such as BERT [35] and word2vec [36]. These embeddings capitalize on the nuanced semantic representations gleaned by these models. The third type involves embeddings extracted from knowledge graphs [37] or online encyclopedias [38], offering additional context and relational insights into the categories.

In the domain of RS, previous studies have predominantly relied on pretrained language models or manually labeled attributes [25], [26], [39]. While these approaches have demonstrated utility, they often fail to capture the entire spectrum of visual characteristics inherent to various categories and demand substantial manual labor. To overcome these challenges, we introduce a pioneering method that uses a multimodal RS network to automatically generate and annotate attributes. This approach is designed to ensure that attributes are not only visually discernible but also exhaustive across all categories, thus amplifying the overall efficacy of the ZSL framework in the context of RS applications.

III. Methodology

A. Problem Statement

ZSL aims to recognize images of unseen classes, denoted as $\mathcal{Y}^U$, by leveraging semantic data extracted from seen classes, denoted as $\mathcal{Y}^S$. It is crucial to note that the sets of seen and unseen classes are mutually exclusive, i.e., $\mathcal{Y}^U \cap \mathcal{Y}^S = \emptyset$. We define the spaces of seen and unseen classes as S and U, respectively, with $\boldsymbol{A}$ representing the attribute vocabulary assigned to both. The comprehensive image space for these classes is represented by $\mathcal{X} = \mathcal{X}^S \cup \mathcal{X}^U$. In ZSL, training data are exclusively derived from seen classes, where each sample is paired with its corresponding class attributes. This is formalized as $\mathcal{D}^S = \{(\mathbf{X}, Y \mid \mathbf{X} \in \mathcal{X}^S, Y \in \mathcal{Y}^S\}$. In this context, x represents an image from the seen class image space $\mathcal{X}^S$, while Y represents the class label. During the training phase, attribute vectors for both seen and unseen classes are available. In the GZSL scenario, the goal is expanded to include classifying images from both seen and unseen categories, which can be expressed as $\mathcal{X} \rightarrow \mathcal{Y}^S \cup \mathcal{Y}^U$.

B. Overall Framework

As illustrated in Fig. 2, we proposed a robust and discriminative framework designed to align vision and semantic spaces in the context of zero-shot RS scene classification. Our proposed method encompasses four critical components.

1) A backbone network that leverages a ViT architecture equipped with self-attention layers to extract image features, which serve as the foundational representation for each image.
2) A classifier with vision attribute mapping that predicts class labels for each image. This is achieved by assessing attribute similarity and utilizing an attention module to direct the localization of local features.
3) The integration of CBCL to bolster the discriminative capacity of local features. This is accomplished within an attribute-guided supervised learning framework that aligns feature representations based on class-attribute relationships.
4) The introduction of a mean-teacher framework to fortify the robustness of the RDVSA model. This framework harnesses self-supervised signals from images subjected

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to various perturbations, thereby enhancing resistance to irrelevant attribute regions. Following comprehensive experimental comparisons, the classification output produced by the student model has been chosen to serve as the predictive classification scores for the ZSL model.

C. Automatic Annotation of Attributes Using CLIP

The CLIP model was initially pretrained on a vast dataset of general optical images sourced from the Internet, which, unfortunately, lacks the unique characteristics inherent to RS imagery. This domain-irrelevant content results in discrepancies that hinder the model's performance on RS tasks. To address these challenges, we fine-tuned the CLIP model in accordance with the methodology presented in [14], employing RS scene images coupled with descriptions from the RSICD dataset. We incorporated an attribute vocabulary $\boldsymbol{A}$, encompassing K attributes $\{\mathbf{a_1}, \mathbf{a_2}, \ldots, \mathbf{a_K}\}$. The CLIP model was subsequently calibrated to align with this attribute vocabulary, thereby enabling effective attribute-based annotation and enhancing its applicability to RS tasks.

Given an RS scene class Y and attribute $\mathbf{a} \in \boldsymbol{A}$, the objective is to assign an attribute value $v_a(Y) \in R$, which signifies the likelihood of attribute $\mathbf{a}$ being present within class Y. The CLIP model facilitates the assessment of the association strength $v_a(Y)$ between attribute $\mathbf{a}$ and the class Y. In addition, the CLIP model comprises a semantic encoder $E_t(\cdot)$ and a visual encoder $E_v(\cdot)$. These encoders project attribute names and detected images into a common embedding space. The real-valued confidence $v_a(Y)$ is derived from a similarity measure f_{sim} as follows:

$$v_a(Y) = \sum_{i=1}^{m} f_{\text{sim}}(E_t(\mathbf{a}), E_v(\mathbf{X})), \tag{1}$$

where the similarity measure f_{sim} is computed as the dot product

$$f_{\text{sim}}(\phi, \varphi) = \langle \phi, \varphi \rangle. \tag{2}$$

Consequently, this methodology yields a class embedding that encapsulates all attribute values associated with each category. The framework obviates the requirement for manually annotating individual images with each attribute, opting instead to designate $\boldsymbol{v_A}(Y)$ as the representative attribute vector for each class.

D. Vision Attribute Mapping for Classification

We proposed an end-to-end learning framework for vision attribute mapping. Specifically, we utilize the widely adopted ViT as our backbone network for feature extraction. An input image $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, where H, W, and C are the height, width, and number of channels, respectively, is initially divided into a series of smaller patches, denoted as $\{\mathbf{X_n}\}_{n=1}^{N}$. Each patch $\mathbf{X_n}$ has dimensions $\mathbb{R}^{(H/l)\times(W/l)\times C}$, where l is the patch size, and the total number of patches is given by $N = l \times l$. These patches are mapped into D-dimensinoal embedding vectors via a linear embedding layer, resulting in the initial embeddings $\mathbf{Z^0} \in R^{N \times D}$. The image patches are processed through a Transformer encoder, which is composed of L layers, each containing multihead self-attention (MHSA) and multilayer perceptrons (MLPs). This structure produces an output representation $\mathbf{Z^L} \in \mathbb{R}^{N \times D}$.

To capture the relationship between visual features and semantic attributes, we introduce a mapping layer that learns a set of K prototype vectors, denoted as $\mathbf{p_1}, \mathbf{p_2}, \ldots, \mathbf{p_K}$. Each prototype $\mathbf{p_i} \in \mathbb{R}^D$ encodes the visual characteristics associated with the ith attribute $\mathbf{a_i}$, while each local embedding $\mathbf{z_n} \in \mathbb{R}^D$ captures information from a local image patch $\mathbf{X_n}$. To ensure that each prototype vector accurately reflects the visual semantics of its corresponding attribute, we compute the similarity between $\mathbf{p_i}$ and each local feature $\mathbf{z_n}$ using a dot product. This similarity, defined as $\text{sim}(\mathbf{p_i}, \mathbf{z_n}) = \mathbf{p_i} \cdot \mathbf{z_n}$, indicates how likely the region $\mathbf{X_n}$ corresponds to attribute $\mathbf{a_i}$. The resulting similarity scores are then reshaped into an attention map $\mathbf{M_i} \in \mathbb{R}^{l \times l}$, which represents the spatial probability of attribute $\mathbf{a_i}$ appearing in the entire image $\mathbf{X}$. The following describes the procedure for constructing the attention map $\mathbf{M_i}$:

$$\mathbf{M_i} = \{\mathbf{p_i} \cdot \mathbf{z_n}\}_{n=1}^{N}. \tag{3}$$

To estimate the attribute value $v_a(\mathbf{X})$, we compute the maximum similarity between each image patch and its corresponding attribute prototype, as given by

$$v_a(\mathbf{X}) = \max_n \{\mathbf{z_n} \cdot \mathbf{p_i}\}, \tag{4}$$

where $v_a(\mathbf{X}) \in R$ represents the predicted attribute value. In general, the complete attribute embedding for an image is represented as $\boldsymbol{v_A}(\mathbf{X}) \in R^K$. For classification, the compatibility score is computed by evaluating the similarity between the predicted attribute embedding and the ground-truth embedding of each training class

$$S(\mathbf{X}, Y) = \boldsymbol{v_A}(\mathbf{X}) \cdot \boldsymbol{v_A}(Y). \tag{5}$$

To optimize this alignment, we apply cross-entropy loss, which encourages the predicted attribute embeddings to closely match the true attribute labels by maximizing compatibility between the image and its respective attribute values. The loss is defined as follows:

$$L_{sc}(\mathbf{X}) = -\log \frac{\exp[S(\mathbf{X}, Y)]}{\sum_{Y_i \in \mathcal{Y}^s} \exp[S(\mathbf{X}, Y_i)]}. \tag{6}$$

We apply a cross-entropy loss to enhance the compatibility score between an image $\mathbf{X}$ and its corresponding label Y while reducing scores for nonmatching image–label pairs. This promotes clearer class separation and strengthens model performance. In addition to cross-entropy loss, we incorporate the widely used attribute regression loss [10] to further enhance the learning of attribute prototypes and the backbone network. Since accurate attribute localization is critical for ZSL, we approach it as a regression problem. To provide effective supervision, we minimize the mean squared error (mse) loss, formulated as

$$L_{\text{mse}}(\mathbf{X}) = \|\boldsymbol{v_A}(\mathbf{X}) - \boldsymbol{v_A}(Y)\|_2. \tag{7}$$

By minimizing this regression loss, we ensure that the learned image representations capture both attribute information and visual characteristics, enhancing the overall model's alignment between image features and attributes.

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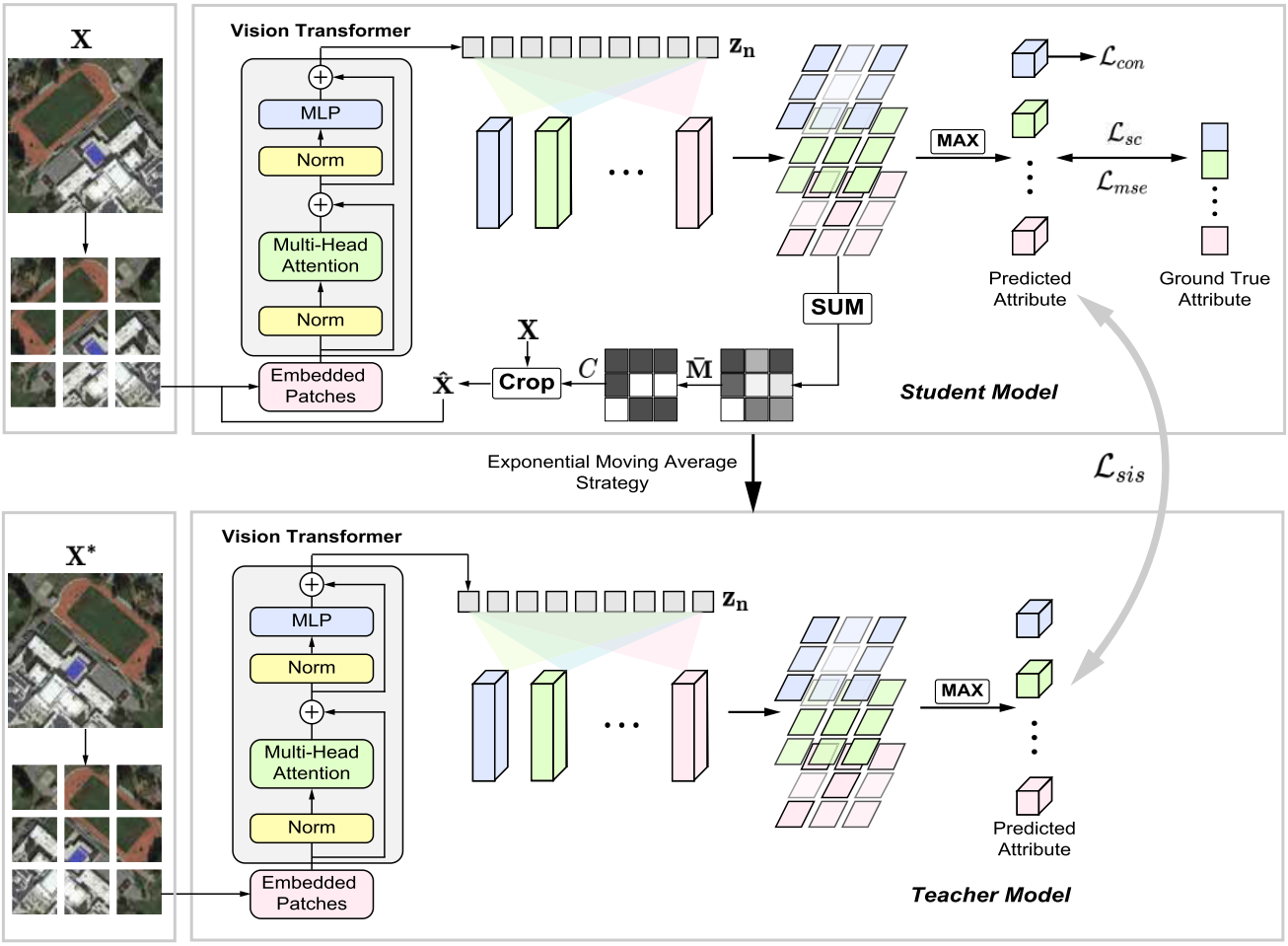


Fig. 2. Framework of the proposed RDVSA. The student model (top) and the teacher model (bottom) both take augmented satellite images $\mathbf{X}$ and $\mathbf{X}^*$ as inputs, respectively. Each image is divided into N patches $\{\mathbf{X_n}\}_{n=1}^{N}$, embedded and passed through a ViT to produce patch-level features $\mathbf{z_n}$. The student model computes attention maps $\{\mathbf{M_i}\}_{i=1}^{K}$ to guide patch-level cropping, producing $\hat{\mathbf{X}}$. The sum of patch representations and their attention maps yields predicted attributes, compared against ground truth via three loss functions: semantic compatibility $\mathcal{L}_{\text{sc}}$, attribute regression $\mathcal{L}_{\text{mse}}$, and contrastive CBCL loss $\mathcal{L}_{\text{con}}$. The teacher model processes $\mathbf{X}^*$ to provide supervision via a consistency loss $\mathcal{L}_{\text{sis}}$. Teacher parameters are updated using an EMA strategy. During testing, the student model predicts attributes and matches them to unseen classes.

E. Attention Module

To leverage the attention maps $\mathbf{m_i}$, we incorporate an attention focusing module that highlights important regions of the image. Specifically, given K attribute-based attention maps $\{\mathbf{M_i}\}_{i=1}^{K}$, we first aggregate these maps by computing the mean attention map, which is defined as follows:

$$\bar{\mathbf{M}} = \frac{1}{K}\sum_{i=1}^{K}\mathbf{M_i}. \tag{8}$$

We then compute the average attention value within $\bar{\mathbf{M}}$ as follows:

$$\bar{m} = \frac{1}{N}\sum_{\alpha=1}^{l}\sum_{\beta=1}^{l}\bar{\mathbf{M}}_{\alpha,\beta}, \tag{9}$$

where α and β refer to spatial coordinates and $N = l \times l$ represents the number of spatial locations. Using $\bar{m}$ as a threshold, we create a binary concentration mask C of size $l \times l$, which identifies the most informative areas of the image

$$C_{\alpha,\beta} = \begin{cases} 1, & \text{if } \bar{\mathbf{M}}_{\alpha,\beta} \geq \bar{m} \\ 0, & \text{if } \bar{\mathbf{M}}_{\alpha,\beta} < \bar{m} \end{cases}, \tag{10}$$

where pixels within $\bar{\mathbf{M}}$ that have attention values above the mean $\bar{m}$ are assigned a value of 1, whereas those below the mean are set to 0. Subsequently, we identify the smallest bounding box that encompasses all regions marked with a value of 1 in C. This bounding box is then used to crop the original image $\mathbf{X}$, resulting in a focused image $\hat{\mathbf{X}}$. This cropped image, $\hat{\mathbf{X}}$, is reintroduced to the ViT model for further classification. During training, we alternate between $\mathbf{X}$ and $\hat{\mathbf{X}}$ as inputs, which encourages the model to sharpen its focus on the most pertinent regions, thereby enhancing classification accuracy.

F. CBCL for Visual-Semantic Feature Alignment

Traditional attribute networks bolster visual-semantic alignment by regressing attributes from local region features, thereby enhancing the generalization capabilities of ZSL models. However, these methods predominantly concentrate on embedding semantic attributes, often neglecting the discriminative quality of the extracted features. To augment the discriminative potency of features gleaned by attribute-guided backbones, we introduce CBCL. This method ensures that the contrastive learning process zeroes in on the most

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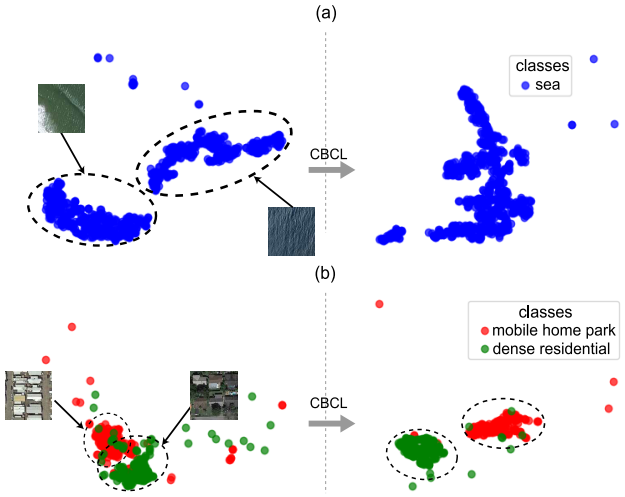


Fig. 3. CBCL. t-SNE embeddings for (a) intraclass diversity and (b) interclass confusion.

representative features of each class, thereby sharpening class distinction and bolstering the model's discriminative ability.

In the context of RS imagery, two significant challenges present themselves. The first is intraclass diversity. As illustrated in Fig. 3(a), the appearance of "sea" can vary significantly across different environmental conditions. In some instances, the sea may appear green, while in others, it might appear blue. This variation leads to the dispersion of feature vectors for the same class across various regions. CBCL addresses this issue by emphasizing the most consistent features within each class, diminishing the impact of background diversity, and thereby enhancing the model's robustness. The second challenge is interclass confusion. Classes such as "mobile home park" and "dense residential" often share similar visual characteristics, like "buildings" and "roads," as depicted in Fig. 3(b). Traditional contrastive learning, which relies on the random selection of positive and negative samples, may struggle to differentiate between such closely related classes effectively. By employing CBCL, our approach ensures a more targeted learning process that effectively tackles both interclass confusion and intraclass diversity.

For each input image $\mathbf{X_i}$, after processing through the backbone ViT and the visual-attribute mapping layer, we obtain the predicted attribute vector output $\boldsymbol{v}_A(\mathbf{X_i})$. The class centroid $\boldsymbol{v}_c$ is then defined as the mean of the predicted attribute vectors for all images within that class

$$\boldsymbol{v}_c = \frac{1}{m}\sum_{i=1}^{m} \boldsymbol{v}_A(\mathbf{X_i}), \tag{11}$$

where m represents the number of images in that class. We construct the predicted attribute of positive samples $\{\boldsymbol{v}_A(\mathbf{X_i}), \boldsymbol{v}_c^+\}$ and negative samples $\{\boldsymbol{v}_{c_1^-}, \ldots, \boldsymbol{v}_{c_n^-}\}$, where n is the number of exclusive categories. The positive sample $\boldsymbol{v}_c^+$ is the centroid of the corresponding category and negative samples $\{\boldsymbol{v}_{c_1^-}, \ldots, \boldsymbol{v}_{c_n^-}\}$ are derived from the centroids of mutually exclusive categories. Then, similarity score is calculated for the predicted attribute f_A and a centroid f_c as follows:

$$\Psi(\boldsymbol{v}_A(\mathbf{X_i}), \boldsymbol{v}_c) = \exp\left(\boldsymbol{v}_A(\mathbf{X_i})^\top \boldsymbol{v}_c/\tau\right), \tag{12}$$

where τ is the temperature coefficient of the contrastive learning. The attribute contrastive embedding can be formulated as follows:

$$\mathcal{L}_{\text{con}} = -\log \frac{\Psi\left(\boldsymbol{v}_A, \boldsymbol{v}_c^+\right)}{\Psi\left(\boldsymbol{v}_A, \boldsymbol{v}_c^+\right) + \sum_{i=1}^{n} \Psi\left(\boldsymbol{v}_A, \boldsymbol{v}_{c_i^-}\right)}. \tag{13}$$

G. Consistency Regularization

To bolster the robustness of cross-modal semantic alignment within the embedding space, we employ the mean-teacher framework. This strategy enhances model stability by ensuring that predictions are consistent across various augmentations of the same image. The approach is based on a student–teacher model configuration, where both models have identical architectures but differ in their parameter update mechanisms. The teacher model's parameters are refined by applying an exponential moving average (EMA) to the student model's parameters, which is governed by the following formula:

$$\theta_t = \alpha\theta_{t-1} + (1-\alpha)\theta_s, \tag{14}$$

where α serves as the smoothing factor. The student and teacher models process separate augmentations of the same input image, denoted as $\mathbf{X}$ and $\mathbf{X}^*$, respectively. These images pass through identical networks with different parameters, yielding classification scores $S(\mathbf{X}, Y)$ and $S(\mathbf{X}^*, Y)$. We enforce consistency between the student and teacher models using mean square error as the regularization term

$$\mathcal{L}_{\text{sis}} = \left\| S(\mathbf{X}, Y) - S\left(\mathbf{X}^*, Y\right) \right\|. \tag{15}$$

We adopted the student model's outputs as the conclusive classification scores. By harnessing the mean-teacher framework, we have crafted a model that exhibits greater consistency and resilience when subjected to a variety of image augmentations.

H. Overall Loss Function

Finally, we optimize the model by minimizing the following comprehensive loss function:

$$\mathcal{L} = \mathcal{L}_{\text{sc}} + \lambda_1\mathcal{L}_{\text{con}} + \lambda_2\mathcal{L}_{\text{mse}} + \lambda_3\mathcal{L}_{\text{sis}}. \tag{16}$$

The overall loss function is composed of four terms, balanced by the hyperparameters λ_1–λ_3. In each batch, our network jointly optimizes the transformer and classification layers using the combined loss function.

I. Training and Testing Procedure

To clearly describe the training and testing process of the RD-VSA model, we summarize the procedure in Algorithm 1. The algorithm includes key steps such as feature extraction using the ViT backbone, attention map computation, and generation of cropped images based on attention. The model is trained by minimizing a combined loss, which includes

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Algorithm 1 Training and Testing Process of RDVSA

Input:
- Training set $\{(\mathbf{X}, Y) \mid \mathbf{X} \in \mathcal{X}^S, Y \in \mathcal{Y}^S\}$.
- Maximum epochs N, patience$\mathcal{L}$, $\Delta\mathcal{L}$, $\mathcal{L}_{\min} \leftarrow \infty$.

Output: Predicted labels $\hat{Y}$ for unseen test samples.
Initialize: ViT backbone and $\boldsymbol{v}_A(y)$.
Training:
for epoch $= 1$**to**N **do**
  $\mathbf{X}$ is divided into N patches, denoted as $\{\mathbf{X_n}\}_{n=1}^{N}$.
  Extract image features: $\mathbf{z_n} = \text{ViT}(\mathbf{X_n})$.
  Compute attention maps $\{\mathbf{M_i}\}_{i=1}^{K}$ via Eq. (3).
  Generate cropped image $\hat{\mathbf{X}}$ via Eqs. (8)–(10).
  Process augmented image $\mathbf{X}^*$ with teacher ViT: $\mathbf{Z_n^*}$.
  Compute $\mathcal{L}_{\text{sc}}$, $\mathcal{L}_{\text{mse}}$, $\mathcal{L}_{\text{con}}$, $\mathcal{L}_{\text{sis}}$ via Eq. (6), (7), (13), (15).
  Compute total loss $\mathcal{L}$ via Eq. (16).
  if $\mathcal{L} < \mathcal{L}_{\min} - \Delta\mathcal{L}$
    then $\mathcal{L}_{\min} = \mathcal{L}, c = 0$.
    Update parameters of student and teacher models.
  else $c = c + 1$
  if $c \geq$ patience$\mathcal{L}$ **then break**.
end for
Testing:
for each unseen test image $\mathbf{X^u} \in \mathcal{X}^U$ **do**
  $\mathbf{X^u}$ is divided into $\{\mathbf{X_n^u}\}_{n=1}^{N}$.
  Extract image features: $\mathbf{z_n} = \text{ViT}(\mathbf{X_n^u})$.
  Predict attribute vector $\boldsymbol{v}_A(\mathbf{X^u})$ via Eq. (4).
  Compute $S(\mathbf{X^u}, Y)$ for all $Y \in \mathcal{Y}^U$ via Eq. (5).
  Predict label: $\hat{Y} = \arg\max_{Y \in \mathcal{Y}^U} S(\mathbf{X^u}, Y)$.
end for

semantic compatibility, attribute regression, CBCL, and consistency loss. The teacher model is updated using the EMA strategy. An early stopping mechanism is applied: training stops if the total loss $\mathcal{L}$ does not improve beyond a margin $\Delta\mathcal{L}$ for a certain number of epochs (patience$\mathcal{L}$). During testing, the model predicts attribute vectors for unseen images and classifies them by comparing with ground-truth attribute embeddings.

IV. Experiment

This section begins with a detailed overview of the dataset and the experimental setup. Following this, we compare our model's results against other SOTA models in both ZSL and GZSL tasks to benchmark its performance. We then perform an ablation study to evaluate the contribution of each component in our model. Finally, we analyze the model's classification accuracy using a confusion matrix and examine the clustering of the generated attribute vectors through a t-SNE visualization.

A. Experiment Settings

1) Dataset: We conduct our ZSL experiments using the well-established large-scale benchmark dataset RSSDIVCS [27], which aggregates scene images from four separate sources: UCM [40], AID [41], NWPU-RESISC45 [42], and RSICB256 [43]. This comprehensive dataset consists of 56 000 images across 70 unique categories, representing a wide variety of environments, from natural landscapes like lakes, mountains, and sea ice to human-impacted areas such as schools, stadiums, and power plants. RSSDIVCS poses significant challenges due to the fine-grained details within RS scenes, where similar land types may appear across a range of unique contexts.

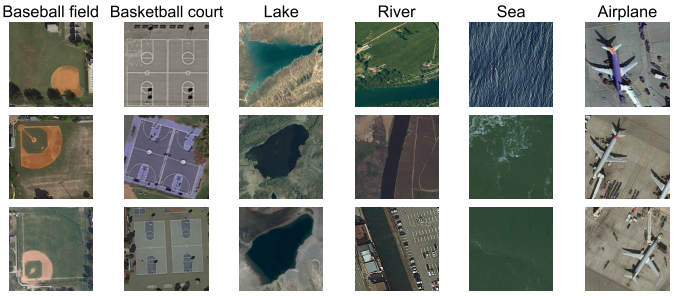


Fig. 4. Examples from the RSSDIVCS dataset.

In Fig. 4, we present some samples from six groups within these fine-grained scene categories. To ensure a fair comparison, we follow the category split approach used in prior research, dividing the 70 classes into seen and unseen subsets with three different splits: 60/10, 50/20, and 40/30. For example, in the 60/10 split, 60 classes serve as seen classes, while the remaining 10 are designated as unseen classes.

2) Metrics: Following standard evaluation protocols, we evaluate the proposed approach by gauging the average per-class Top-1 accuracy (T1) across both traditional ZSL and GZSL tasks. In the context of ZSL, the assessment is centered on T1 solely on a test set comprised of unseen classes. For GZSL, the test set encompasses both seen and unseen classes, providing a more holistic measure of performance. As detailed in [6] and [44], we present two metrics, S and U, which represent the average per-class top-1 accuracy for seen and unseen test classes, respectively, within GZSL scenarios. To assess the overall performance in GZSL, we also calculate the harmonic mean (H) between S and U, defined as follows:

$$H = \frac{2 \times S \times U}{S + U}. \tag{17}$$

This harmonic mean provides a balanced assessment of the model's ability to generalize across both seen and unseen classes.

3) Training Details: For attribute extraction, we utilize a fine-tuned CLIP model to automatically generate attribute annotations. In training our model, we adopt a ViT-B/32 with 12 layers as the primary backbone. The network is optimized end-to-end using the Adam optimizer with $\beta_1 = 0.5$ and $\beta_2 = 0.999$. To initiate training, we freeze the transformer backbone and pretrain the classification module for four epochs at a learning rate of 1×10^{-4} as a warm-up step. Afterward, we unfreeze the full network and continue training for an additional 26 epochs, this time applying a reduced learning rate of 1×10^{-6}.

To make a fair comparison with most existing methods, we also use ResNet18 [45] as an alternative backbone. The

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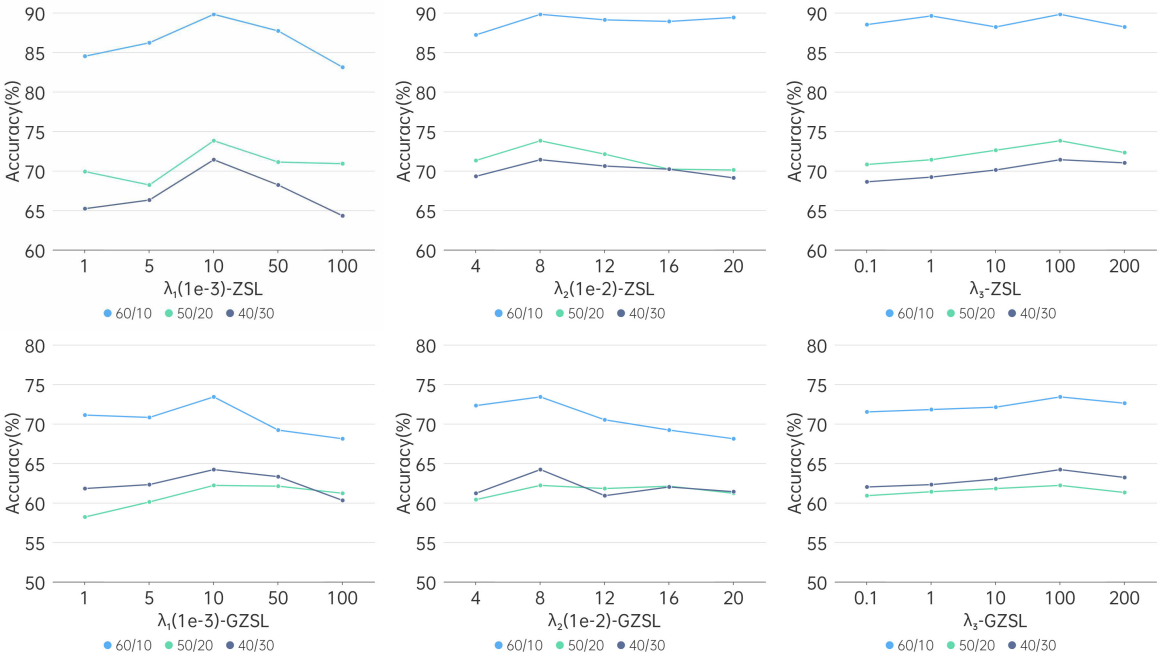


Fig. 5. Analysis of the value of hyperparameters λ_1–λ_3 on three splits.

network is trained with the Adam optimizer, where $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and the learning rate is initially set to 1×10^{-4} with a weight decay of 1×10^{-5}. After the initial warm-up phase (four epochs), we proceed with further training for 26 epochs, reducing the learning rate to 1×10^{-6} to fine-tune the network, ensuring effective feature extraction across both backbones.

4) Hyperparameter Analysis: The selection of hyperparameters λ_1–λ_3 was guided by comprehensive ablation studies and a systematic analysis of their impact on model performance, as demonstrated in Fig. 5 generated from the experimental data.

The final selected weights, $\lambda_1 = 10 \times 1e^{-3}$, $\lambda_2 = 8 \times 1e^{-2}$, and $\lambda_3 = 100$, were chosen to maximize the harmonic mean H across splits while maintaining a balance between the tasks. Specifically, $\lambda_1 = 10 \times 1e^{-3}$ provided the best performance for contrastive loss, enhancing interclass separation and intraclass compactness. Increasing λ_1 beyond 10 led to over-regularization, resulting in a performance decline. For regression loss, $\lambda_2 = 8 \times 1e^{-2}$ struck the ideal balance, ensuring precise attribute alignment without overshadowing other objectives, whereas values higher than eight led to diminishing returns or slight performance declines. As for consistency loss, while $\lambda_3 = 100$ produced the most robust results, even smaller values like $\lambda_3 = 0.1$ still yielded competitive outcomes. This suggests that overprioritizing consistency regularization could prevent the model from capturing meaningful distinctions between samples, which negatively impacts the performance. Overall, the selected hyperparameter values were carefully calibrated to achieve optimal performance while avoiding both overfitting and underfitting.

B. Ablation Study

Our complete loss function comprises four components: $\mathcal{L}_{sc}$ for semantic compatibility, $\mathcal{L}_{mse}$ for attribute regression, $\mathcal{L}_{sis}$ for consistency regularization, and $\mathcal{L}_{con}$ for contrastive learning. To systematically evaluate their contributions, we design five configurations: 1) a baseline with only $\mathcal{L}_{sc}$; 2) baseline plus $\mathcal{L}_{con}$; 3) baseline plus $\mathcal{L}_{mse}$; 4) baseline with $\mathcal{L}_{mse}$ and $\mathcal{L}_{sis}$; and 5) the full model integrating all four losses. Each configuration is tested on 60/10, 50/20, and 40/30 splits under both ZSL and GZSL protocols.

As shown in Table I, the baseline model with $\mathcal{L}_{sc}$ alone achieves moderate performance, serving as the starting point for comparison. Introducing $\mathcal{L}_{con}$ to the baseline enhances the model's ability to separate interclass features but reveals a critical limitation: while unseen class accuracy (U) improves by 20.5% on the 60/10 split, seen class accuracy (S) remains excessively high at 86.3%, resulting in a suboptimal harmonic mean (H) of 56.7%. This indicates that contrastive learning alone amplifies feature discriminability at the cost of overfitting to seen categories. Replacing $\mathcal{L}_{con}$ with $\mathcal{L}_{mse}$ addresses this imbalance by enforcing semantic alignment between visual and attribute embeddings. On the 60/10 split, this configuration boosts the T1 accuracy by 7.4% (from 75.7% to 83.1%) and harmonic mean by 31.3% (from 35.0% to 66.3%) while reducing seen class overfitting (S drops from 90.1% to 74.2%).

Further integrating $\mathcal{L}_{sis}$ with $\mathcal{L}_{mse}$ introduces consistency regularization, which stabilizes feature learning against input perturbations. This refinement improves T1 accuracy by 3.3% on the 50/20 split (from 65.0% to 68.3%) and harmonic mean by 3.2% (from 56.9% to 60.1%), demonstrating enhanced robustness. The full model uniting all components achieves peak performance, where $\mathcal{L}_{con}$ contributes an 11.1% T1 gain on the 40/30 split (from 60.3% to 71.4%) through refined class separation, while $\mathcal{L}_{sis}$ ensures prediction consistency across augmented views. Under GZSL, the full model reduces seen class overfitting by 15.6% (from 90.1% to 74.5% on 60/10) while elevating unseen class accuracy by 50.7% (from 21.7% to 72.4%). The harmonic mean improves by 38.4%, 35.9%,

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TABLE I

IN THE ABLATION STUDY OF OUR PROPOSED MODEL, WE REPORT THE TOP-1 (T1) ACCURACY ON UNSEEN CLASSES FOR ZSL, AS WELL AS THE TOP-1 ACCURACY ON UNSEEN (U), SEEN (S) CLASSES, AND THE HARMONIC MEAN (H) FOR GZSL. OUR MODEL IS TRAINED ONLY WITH THE LOSS $\mathcal{L}_{sc}$ AS THE BASELINE. THE BEST RESULTS ARE IN RED

$\mathcal{L}_{sc}$	$\mathcal{L}_{mse}$	$\mathcal{L}_{sis}$	$\mathcal{L}_{con}$	60/10				50/20				40/30			
				T1	S	U	H	T1	S	U	H	T1	S	U	H
✓	✗	✗	✗	75.7	90.1	21.7	35.0	53.4	74.0	16.0	26.3	51.0	82.7	23.1	36.1
✓	✗	✗	✓	78.2	86.3	42.2	56.7	59.6	68.1	40.4	50.1	53.8	66.9	35.4	46.3
✓	✓	✗	✗	83.1	74.2	60.0	66.3	65.0	61.4	53.1	56.9	59.6	56.3	47.4	51.5
✓	✓	✓	✗	83.2	74.2	66.2	70.0	68.3	67.4	54.3	60.1	60.3	65.6	44.5	53.0
✓	✓	✓	✓	89.8	74.5	72.4	73.4	73.8	70.8	55.5	62.2	71.4	67.8	60.9	64.2

TABLE II

WE COMPARE THE ZSL AND GZSL OF OUR RDVSA MODEL WITH NINE SOTA MODELS, WHERE THE MODELS ARE TRAINED WITH CLASS EMBEDDINGS SR-RSKG. WE REPORT THE TOP-1 ACCURACY (T1) OF ZSL AND THE HARMONIC MEAN (H) OF GZSL. THE BEST RESULTS ARE IN RED

Method (SR-RSKG)	ZSL Accuracy			GZSL Accuracy		
	60/10(T1)	50/20(T1)	40/30(T1)	60/10(H)	50/20(H)	40/30(H)
SAE [46]	22.1	12.8	9.2	28.9	23.7	16.9
DMaP [47]	33.1	20.3	12.9	30.1	23.4	16.2
CIZSL [48]	18.2	8.9	7.1	23.7	13.9	8.1
SPLE [49]	28.5	17.2	10.2	-	-	-
CADA-VAE [50]	50.5	39.6	28.2	38.1	32.9	28.1
ZSC-SA [51]	31.3	19.1	13.6	-	-	-
CE-GZSL [17]	53.6	44.7	32.1	42.9	35.9	32.1
TF-VAEGAN [52]	51.5	41.9	30.0	40.1	35.0	29.2
DAN [28]	53.3	45.2	33.4	40.3	34.1	29.6
RDVSA (ResNet)	55.3	45.0	33.9	48.9	35.1	32.2

and 28.1% across splits, validating the synergistic interplay among the four components in our objective function.

C. Comparison With the State-of-the-Art Methods

In this section, we offer an exhaustive comparison of our model with the current state-of-the-art approaches. To thoroughly assess the efficacy of our model, we conduct experiments utilizing various class embedding strategies. Subsequently, we benchmark the proposed model RDVSA against the top-performing models in each class embedding category.

1) Comparison on Different Class Embeddings: We compare our model with two main categories of state-of-the-art ZSL methods. The first category includes nongenerative models that concentrate on mastering projection functions to align image features with class embeddings, such as SPLE [49], DMaP [47], GSER [53], and DSVA [14]. The second category comprises generative models that leverage generative mechanisms such as autoencoders or GANs to synthesize image features for unseen classes. Key models in this category include SAE [46], ZSC-SA [51], CIZSL [48], CE-GZSL [17], CADA-VAE [50], TF-VAEGAN [52], and DAN [28]. We evaluated these models in three types of class embeddings.

1) *SR-RSKG [28]:* This semantic representation framework for RS scenes is derived from knowledge graphs created with input from ten domain experts.
2) *Word2Vec [36]:* Each RS category is represented by a 300-D vector, generated by a Word2Vec model pretrained on the Wikipedia corpus.
3) *BERT [35]:* Domain experts provided concise descriptions for each RS scene category, which were then transformed into 1024-D embeddings using a BERT model, capturing domain-specific information.

As shown in Table II, the proposed model RDVSA, which was trained with SR-RSKG class embeddings and utilizes ResNet18 as its backbone, outperforms other cutting-edge models in both ZSL and GZSL tasks. Notably, our model achieves the highest harmonic mean (H) across all splits, outperforming the second-best model, DAN [28], by a margin of 8.9% in the 60/10 split for GZSL (H). For ZSL (T1) in the same split, our model surpasses DAN by 2.0%. In the 50/20 split, RDVSA outperforms DAN in GZSL (H) by 1.0%, and in the 40/30 split, we achieve a 0.5% improvement in ZSL (T1) and a 2.6% improvement in GZSL (H). These results highlight the superior generalization ability of RDVSA, especially when it comes to balancing the performance between seen and unseen classes.

In Table III, where Word2Vec is used as class embeddings, our RDVSA model continues to outperform other SOTA models. Compared to the second-best model, GSER [53], RDVSA demonstrates substantial improvements, achieving a 6.4% increase in ZSL (T1) and a 7.5% increase in GZSL (H) in the 60/10 split. In the 50/20 split, our model shows a 1.0% improvement in ZSL (T1), and in the 40/30 split, RDVSA surpasses GSER by 5.3% in ZSL (T1) and 1.6% in GZSL (H). These results emphasize that, even when trained with Word2Vec embeddings, RDVSA provides significant improvements in both ZSL and GZSL tasks.

Furthermore, in Table IV, when using BERT as class embeddings, RDVSA again outperforms other SOTA models in both ZSL and GZSL tasks. For example, in the 60/10 split, RDVSA outperforms the best-performing model, DAN [28], by 6.9% in ZSL (T1) and 11.9% in GZSL (H). Additionally, our model achieves the best performance in all three splits, further illustrating the robustness and versatility of RDVSA

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TABLE III

WE COMPARE THE ZSL AND GZSL OF OUR RDVSA MODEL WITH EIGHT SOTA MODELS, WHERE THE MODELS ARE TRAINED WITH CLASS EMBEDDINGS WORD2VEC. WE REPORT THE TOP-1 ACCURACY (T1) OF ZSL AND THE HARMONIC MEAN (H) OF GZSL. THE BEST RESULTS ARE IN RED

Method (Word2Vec)	ZSL Accuracy			GZSL Accuracy		
	60/10(T1)	50/20(T1)	40/30(T1)	60/10(H)	50/20(H)	40/30(H)
SAE [46]	23.5	13.7	9.6	28.0	21.0	17.2
DMaP [47]	26.0	16.7	10.4	28.9	20.3	16.8
CIZSL [48]	20.6	10.6	6.0	25.2	15.7	9.1
SPLE [49]	20.1	13.2	9.8	-	-	-
CADA-VAE [50]	41.4	30.3	21.2	32.9	30.3	26.1
ZSC-SA [51]	26.7	15.2	12.1	-	-	-
DAN [28]	44.3	34.7	24.3	34.1	31.4	25.6
GSER [53]	48.7	36.2	25.5	38.6	33.8	27.5
RDVSA (ResNet)	55.1	37.2	30.8	46.1	33.6	29.1

TABLE IV

WE COMPARE THE ZSL AND GZSL OF OUR RDVSA MODEL WITH SEVEN SOTA MODELS, WHERE THE MODELS ARE TRAINED WITH CLASS EMBEDDINGS BERT. WE REPORT THE TOP-1 ACCURACY (T1) OF ZSL AND THE HARMONIC MEAN (H) OF GZSL. THE BEST RESULTS ARE IN RED

Method(Bert)	ZSL Accuracy			GZSL Accuracy		
	60/10(T1)	50/20(T1)	40/30(T1)	60/10(H)	50/20(H)	40/30(H)
SAE [46]	22.0	12.4	8.8	28.6	21.5	16.7
DMaP [47]	16.4	15.6	10.0	26.6	19.5	16.3
CIZSL [48]	20.4	10.3	6.2	25.0	15.0	8.6
SPLE [49]	19.0	13.2	8.3	-	-	-
CADA-VAE [50]	48.1	37.1	26.3	36.3	31.5	27.1
ZSC-SA [51]	29.3	18.3	13.1	-	-	-
DAN [28]	50.2	43.4	31.5	38.0	31.5	28.2
RDVSA (ResNet)	57.1	43.0	32.9	49.9	38.3	31.4

TABLE V

COMPARING OUR RDVSA MODEL WITH SOTA MODELS. WE REPORT THE TOP-1 ACCURACY (T1) OF UNSEEN CLASSES UNDER THE ZSL SETTING AND THE HARMONIC MEAN (H) OF BOTH SEEN AND UNSEEN CLASSES UNDER THE GZSL SETTING. WE SELECTED THE STATE-OF-THE-ART MODELS THAT PERFORMED BEST UNDER DIFFERENT CLASS EMBEDDINGS FOR COMPARISON. IN CONTRAST TO THE PRECEDING ABLATION STUDIES ON CLASS EMBEDDINGS, WE PRESENT THE SUPERIOR CLASSIFICATION RESULTS ACHIEVED BY OUR MODEL, WHICH EMPLOYS VIT AS THE BACKBONE AND UTILIZES RSMM-ATTRIBUTES AS CLASS EMBEDDINGS. THE BEST RESULTS ARE IN RED

Class Embeddings	Model	ZSL Accuracy			GZSL Accuracy		
		60/10(T1)	50/10(T1)	40/30(T1)	60/10(H)	50/20(H)	40/30(H)
SR-RSKG	CE-GZSL [17]	53.6	44.7	32.1	42.9	35.9	32.1
	DAN [28]	53.3	45.2	33.4	40.3	34.1	29.6
	RDVSA (ResNet)	55.3	45.0	33.9	48.9	35.1	32.2
	RDVSA(ViT)	75.4	48.9	42.5	57.3	35.5	33.7
Word2Vec	DAN [28]	44.3	34.7	24.3	34.1	31.4	25.6
	GSER [53]	48.7	36.2	25.5	38.6	33.8	27.5
	RDVSA (ResNet)	55.1	37.2	30.8	46.1	33.6	29.1
	RDVSA(ViT)	73.2	40.1	31.2	57.1	37.9	32.1
BERT	CADA-VAE [50]	48.1	37.1	26.3	36.3	31.5	27.1
	DAN [28]	50.2	43.4	31.5	38.0	31.5	28.2
	RDVSA (ResNet)	57.1	43.0	32.9	49.9	38.3	31.4
	RDVSA(ViT)	76.5	42.6	37.3	60.9	43.6	38.1
RSMM-Attributes	RDVSA (ResNet)	72.5	52.9	36.8	54.7	42.2	39.6
	DSVA(ViT) [14]	84.0	64.2	60.2	67.7	56.5	49.9
	RDVSA (ViT)	89.9	73.8	71.4	73.4	62.2	64.2

in image classification tasks under various class embeddings. These results highlight the generalization capability of RDVSA across different embedding strategies. By consistently outperforming other SOTA models across ZSL and GZSL tasks with SR-RSKG, Word2Vec, and BERT, our model demonstrates its ability to handle various types of class embeddings, emphasizing its robustness in RS image classification tasks.

2) RSMM-Attributes: To conduct a thorough evaluation, we have equipped our RDVSA model with two distinct backbone architectures: ResNet18 [45] (ResNet) and ViT [52]. Table V details the performance of the top three models utilizing diverse class embeddings. Of particular note is the significant enhancement in our RDVSA model's performance when employing ViT as the backbone and RSMM-Attributes for class embeddings. Our model secures the highest Top-1 accuracy (T1) in the ZSL task and the optimal harmonic mean (H) in GZSL, outperforming all other leading models across both scenarios. For example, in the 60/10 split, our RDVSA model delivers an impressive T1 accuracy of 89.9% and a harmonic mean of 73.4%. These outcomes surpass those of the second-ranked model, DSVA, by 5.9% in ZSL and by

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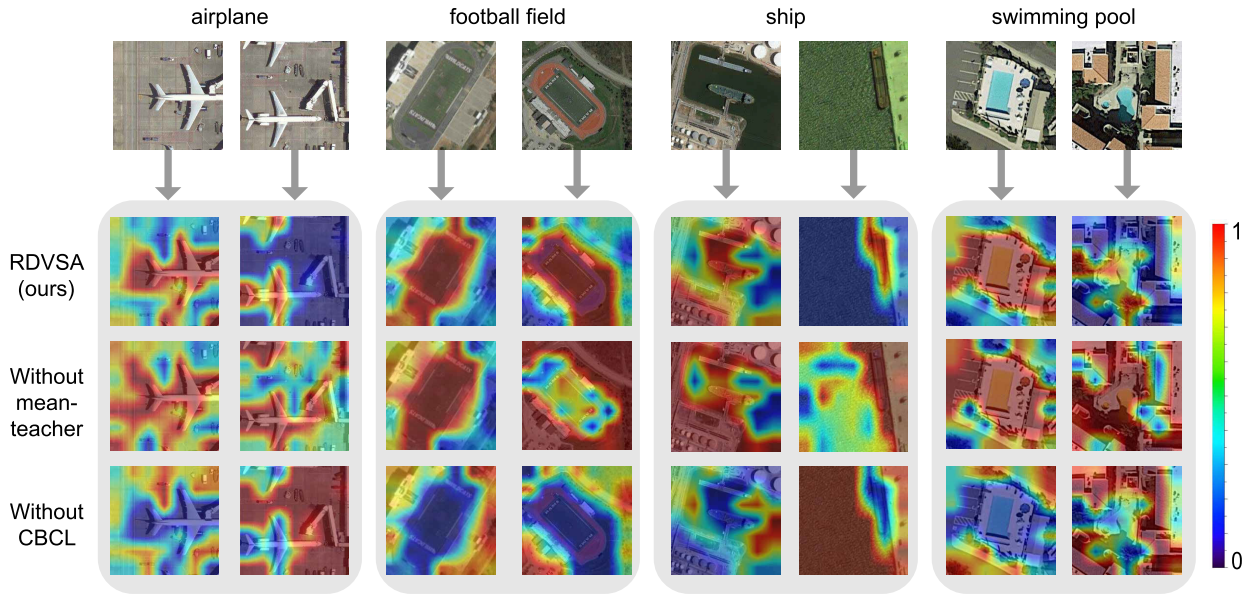


Fig. 6. CAMs for RS scene categories under different model configurations. Each column corresponds to a scene category—airplane, football field, ship, and swimming pool (from left to right). In each column, the top row shows the original input image. The second row shows the CAM generated by the full RDVSA model, including both the mean-teacher framework and the CBCL module. The third row shows the CAM from the RDVSA model without the mean-teacher component, and the fourth row shows the CAM from the RDVSA model without the CBCL module.

5.7% in GZSL, highlighting substantial improvements in both tasks.

In the more challenging 40/30 split, our RDVSA model surpasses the next best model, DVSA, by 11.2% in T1 for ZSL and by 14.3% in H for GZSL. These results can be ascribed, in part, to the efficacy of RSMM-Attributes in capturing more precise visual-semantic correlations, and in part, to the advanced features of the RDVSA model, which includes the integration of CBCL and consistency regularization constraints. These mechanisms empower the model to produce more discriminative and robust visual features and semantic representations, culminating in enhanced visual-semantic alignment. This refined alignment is directly responsible for the exceptional zero-shot classification capabilities exhibited by our model.

D. Qualitative and Visual Analysis

1) CAMs Visualization: To further validate the efficacy of our RDVSA model, we conducted supplementary experiments using class activation maps (CAMs) on four RS scene categories—airplane, football field, ship, and swimming pool—as illustrated in Fig. 6. The image includes the original images and their CAMs under three model variants: full RDVSA model and the CBCL module, RDVSA without the mean-teacher, and RDVSA without CBCL. The top row displays the original input image. The second row presents the CAM produced by the complete RDVSA model. The third row contains the CAM from the RDVSA model without the mean-teacher component, and the fourth row shows the CAM from the RDVSA model without the CBCL module.

Fig. 6 compares the CAMs produced by different versions of the RDVSA model. Adding the mean-teacher framework improves robustness, and adding CBCL module enhances discriminative precision. For example, in the airplane category, the full RDVSA model highlights the entire fuselage and wings. When mean-teacher is removed, CAMs become fuzzy. Parts of the aircraft are missing under different viewpoints. When CBCL is removed, activations spread onto the airport tarmac. This shows that CBCL helps the model ignore irrelevant ground details. Similarly, in the ship category, the full model consistently localizes the hull and deck. Without a mean-teacher, CAMs only cover the bow or stern. This indicates less stable focus across images. Without CBCL, activations extend onto the surrounding water. This reflects reduced discriminative power. The mean-teacher framework ensures stable attention on the whole ship. CBCL keeps the focus from drifting into the background.

Furthermore, for the football field and swimming pool scenes, the pattern is similar. The full model covers the entire playing surface of the football field. Without a mean teacher, CAMs only partially cover the field. Without CBCL, activations appear on the adjacent track or parking area. In the swimming pool example, the full model focuses cleanly on the pool basin. Removing mean-teacher leads to partial coverage of the pool edges. Removing CBCL produces activations on nearby seating or pavement. Overall, the mean-teacher framework makes localization more consistent across conditions. The CBCL module sharpens the model's focus on correct scene elements. Overall, the detailed examination of Fig. 6 confirms that both the mean-teacher framework and the CBCL module contribute significantly to refining the attention mechanisms within the RDVSA model. These enhancements not only improve the precision of the visual focus but also strengthen the overall visual-semantic alignment, thereby bolstering the model's robustness and effectiveness in zero-shot RS scene classification.

2) Confusion Matrix: To thoroughly evaluate the efficacy of CBCL, we present a visualization of the confusion matrix

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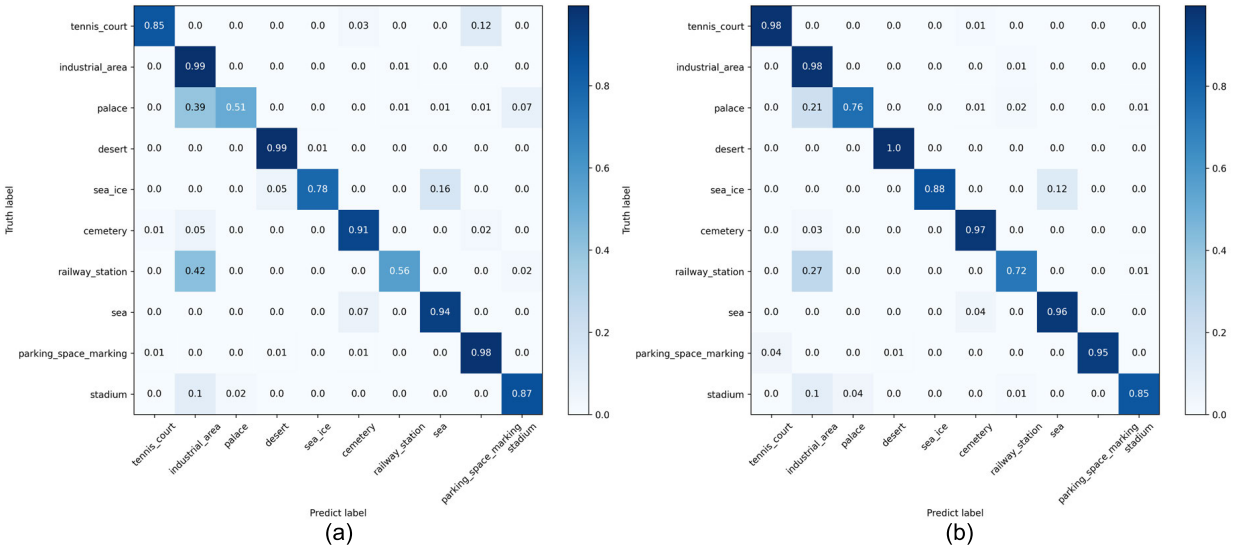


Fig. 7. Confusion matrices in 60/10 split. (a) Confusion matrix for the RDVSA model without CBCL. (b) Confusion matrix for the complete RDVSA model.

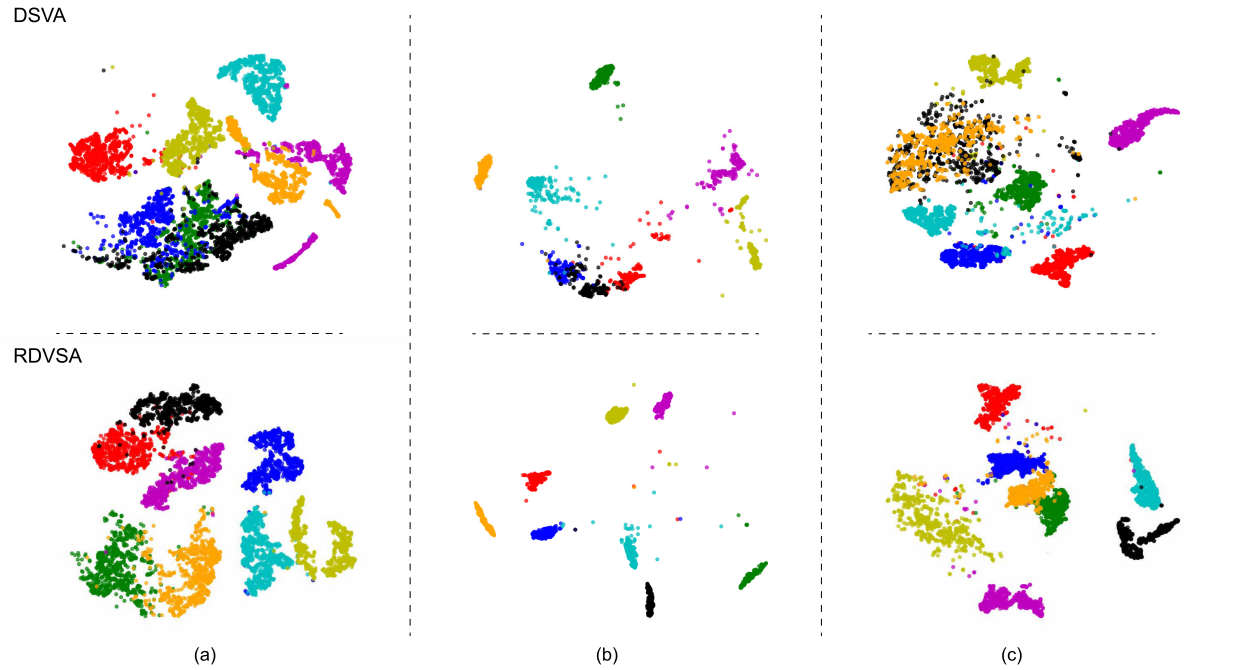


Fig. 8. Attribute visualization by t-SNE embeddings. The top figures illustrate the results of the DSVA model and the bottom figures represent the results of the RDVSA model. (a) Predicted attribute vector on ZSL setting. (b) Predicted attribute vector on GZSL setting (seen classes). (c) Predicted attribute vector on GZSL setting (unseen classes).

based on a 60/10 split dataset, utilizing ViT and RSMM-attribute. The visualization is depicted in Fig. 7. Within the confusion matrix, each column corresponds to the true labels, while each row corresponds to the predicted labels. The values within the matrix represent the proportion of true labels that were categorized under a specific class. The diagonal elements indicate correct classifications, whereas the off-diagonal elements signify incorrect predictions. It is evident that the model without CBCL has lower classification accuracy compared to the comprehensive RDVSA model.

Furthermore, we observed that railway stations and palaces are frequently misclassified as industrial areas. This is attributed to the similarity in both their image features and their corresponding attribute vectors, which complicates the classification task. Without the incorporation of CBCL, a substantial 42% of railway stations and 39% of palaces were incorrectly identified as industrial areas. However, with CBCL, the misclassification rates for railway stations and palaces were significantly reduced to 27% and 21%, respectively. This outcome underscores the enhanced discriminability of attribute vectors generated under CBCL's guidance, particularly in RS contexts characterized by high similarity, where this effect is markedly pronounced.

3) Attribute Visualization: To visually assess the efficacy of our proposed methodology, we generated t-SNE [54] plots to juxtapose the predicted attribute vectors $\boldsymbol{v}_A(\mathbf{X_i})(x)$ yielded by our RDVSA model with those from the DSVA model [14]. During the testing phase, the attribute vectors for the ten

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classes were embedded into a 2-D space for visual representation. Employing a 50/20 split, we crafted t-SNE visualizations depicted in Fig. 8 for both ZSL and GZSL scenarios. For ZSL, we randomly selected eight samples from classes not encountered during training, whereas for GZSL, we selected eight samples from seen classes and eight from unseen classes to exhibit the outcomes.

Notably, the top figures illustrate that for the DSVA model [14], the predicted attribute vectors exhibit heightened dispersion within each class across both ZSL and GZSL settings, characterized by broadened interclass margins and pronounced amalgamation between disparate classes. In contrast, the bottom figures, which represent our proposed RDVSA model, showcase a significant enhancement: the attribute vectors are more tightly clustered within individual classes and exhibit greater interclass separation, thereby bolstering discriminability. Furthermore, a crucial observation is that RDVSA substantially addresses the issue evident in Fig. 8(a), where samples denoted by purple points, belonging to the class "harbor," are dispersed across various regions in the upper figure but are consolidated in the lower figure. This observation underscores that our proposed model effectively diminishes class overlap, establishing more pronounced class boundaries and bolstering intraclass coherence.

In Fig. 8(b) and (c), compared to the top figures, the bottom figures demonstrate a pronounced decrease in the intermingling of projection points from distinct classes, culminating in more distinct and separable clusters. These findings substantiate that our proposed contrastive constraint approach engenders more robust and distinctive attribute vectors, which in turn facilitates effective cross-modal vision-semantic alignment and markedly enhances zero-shot classification accuracy for RS imagery.

V. Conclusion

In this work, we introduce an innovative visual-attribute alignment framework designed to enhance both the discriminative capacity and robustness of ZSL models for RS scene classification. Our approach performs classification by projecting visual images into an attribute space, with a focus on learning localized embeddings that capture semantic attribute regions, thereby preserving the model's generalization capability. To further distinguish between classes, we incorporate contrastive learning to increase the separability of attribute embeddings, establishing clear interclass boundaries. Additionally, our method leverages a self-supervised consistency regularization strategy to strengthen model resilience. Extensive experiments conducted on widely used datasets with diverse class embeddings and backbone architectures underscore the versatility and efficacy of our approach, highlighting its potential for robust zero-shot classification in RS applications.

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